

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document search engine (*Python FastAPI*) for a competitive high school debate research SaaS platform hosted on *AWS*, which drew over **10K+ site visitors**. Collaborated with an *Agile* team of **10 engineers**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**
- Designed and built a text-retrieval algorithm that validates source URLs and filters statistical outliers with **IQR**
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Provided feedback on products in development in a **5-page report** recommending engagement strategies to the CEO
- Tested **3 experimental features** across **7 iterations of UI/UX testing** on an app with **100K+ users**
- Redesigned the onboarding flow on *Figma*, cutting it from **12 to 5 pages**, then validated it with **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **3 years** of TikTok post data (**200+ posts**) with *Python (pandas)*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised **69.3%** of views despite being only **7%** of all posted content
- Recommended a “fun facts” strategy the Executive Director adopted, lifting engagement **24% (8K+ views)**

PROJECTS

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Engineered a smart bin prototype on an *Arduino Uno* that sorts each waste item with **under 2 seconds** of latency
- Trained an *EfficientNetV2-S* model (**4.39M** parameters) in *PyTorch* on **15K+ images** for **96.13% accuracy**
- Published the work as an *arXiv* paper with **2 datasets** on *Kaggle*. **2nd at Arizona State Science Fair**
- Built a synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) producing **9K** contamination images

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a suite of **3 edge AI tools** that collect data on a school’s energy, water and food consumption
- Built a *Python FastAPI* back end that ingests live water, waste & energy data from all **3 tools** for anomaly detection
- Ran **4 Random Forest models** to detect anomalies and delivered actionable steps via a *Gemma* Voice LLM
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Designed a **5-step Python** scheduling algorithm that scans calendars, ranks time slots and emails confirmations
- Performed quality assurance by testing the scheduling algorithm across **14 unit and integration test cases**
- Built a scheduling agent (*React/Firebase*) that auto-books meetings for groups of **30+ people** on Google Calendar
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of **4** to build a pricing algorithm for **Kamion (YC S22)**. **2nd Place at UIUC Founder’s Hackathon**
- Debugged the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Scraped **15,000** truck-listing images and audited data quality by de-duplicating listings and filtering non-USD prices
- Designed a *JSON schema* for a *Claude Sonnet 4.6* agent to extract condition, brand, model and year from photos

SKILLS

Programming Languages: C++, Java, Python, JavaScript, SQL, HTML/CSS

Frameworks/Platforms: AWS, React, Next.js, Node.js, FastAPI, Firebase, Flask, PostgreSQL, MySQL, Supabase

DevTools: Arduino IDE, Clang, Catch2, Git, GitHub, CI/CD, Docker, VS Code, Claude Code, Codex, Figma, Bun

Concepts/Practices: Software Testing, Debugging, Embedded Software, Quality Assurance, Data Structures